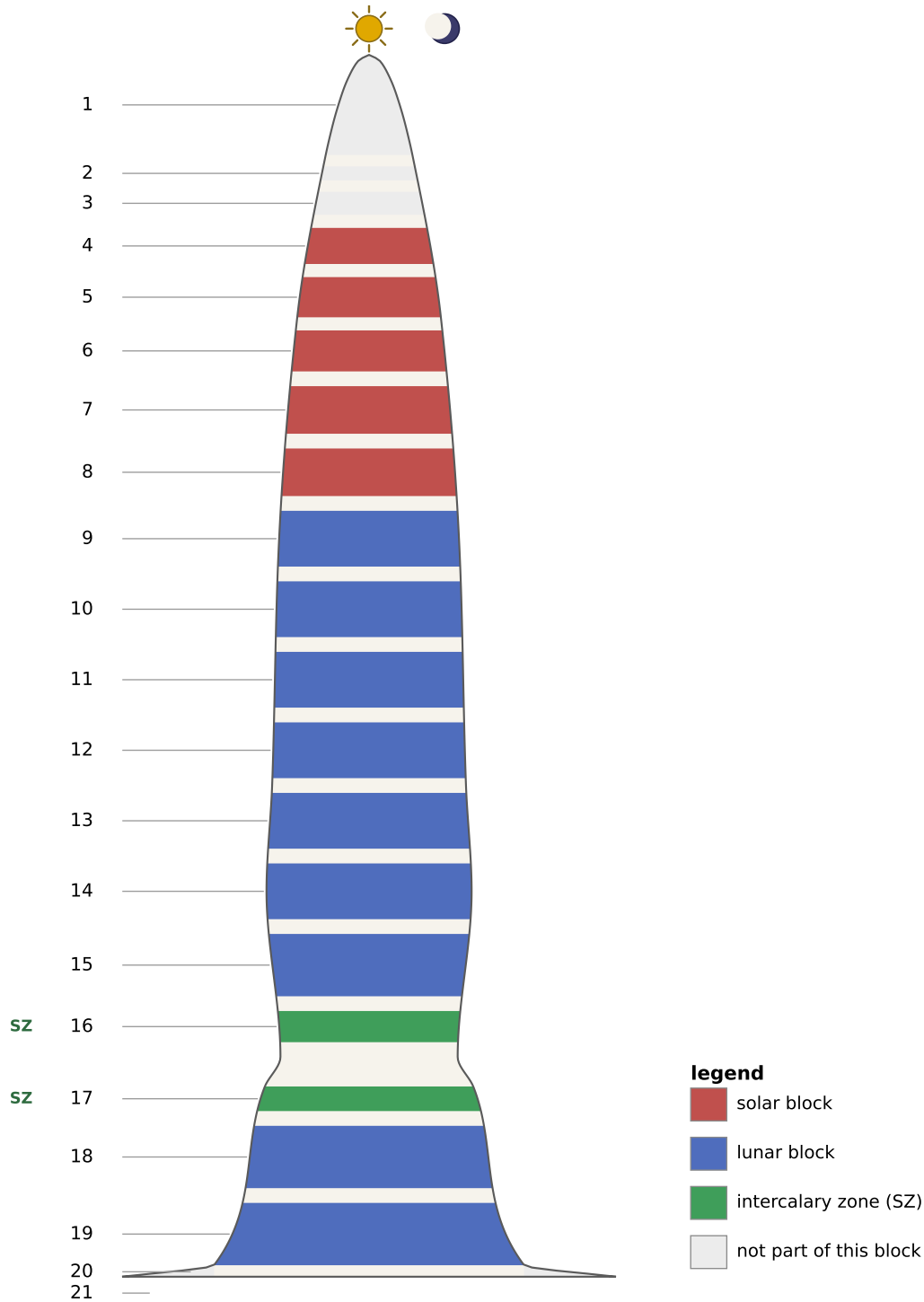
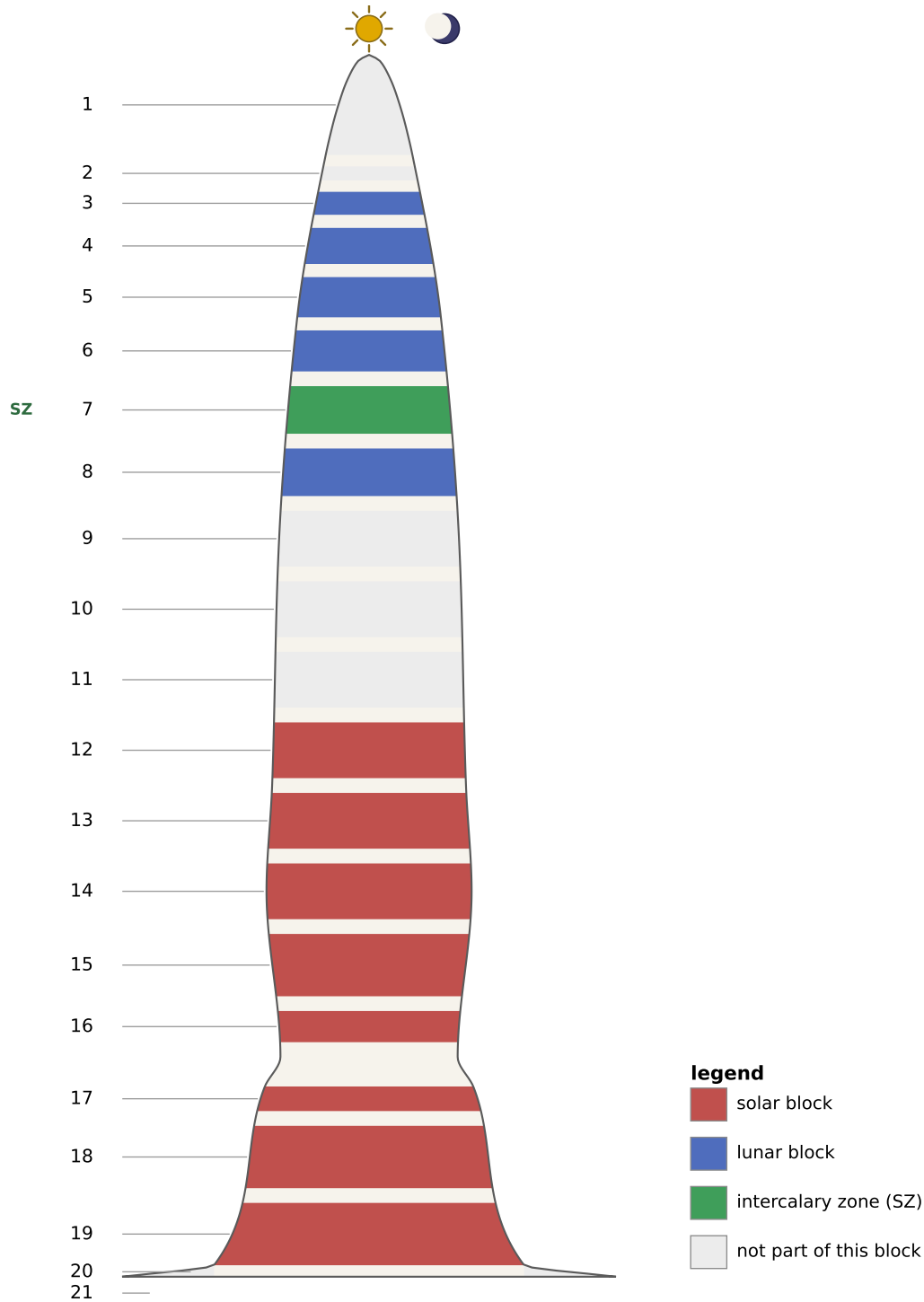
61.1 1 year / 12 lunations (4 QY / 12 L)



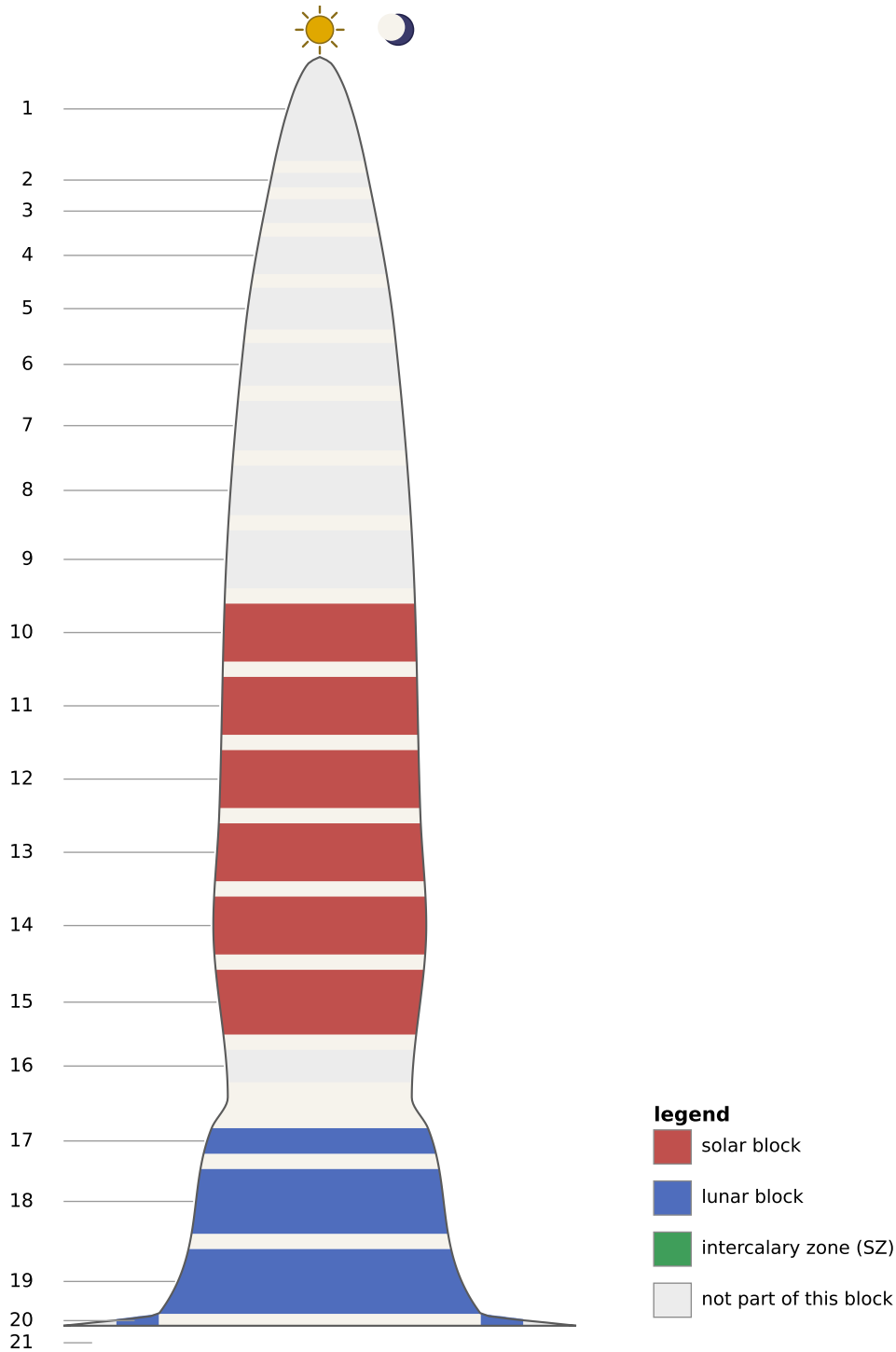
solar 365 d (zones 4-8) | lunar 354 d (zones 15, 18-19) | delta 11 d

61.2 1 year + 1 quarter / 15 lunations (5 QY / 15 L)



delta 16 d | no zone arithmetic in the source: 5/4 J = 458 d vs 15 L = 442 d (absolute), deviates +-2 d

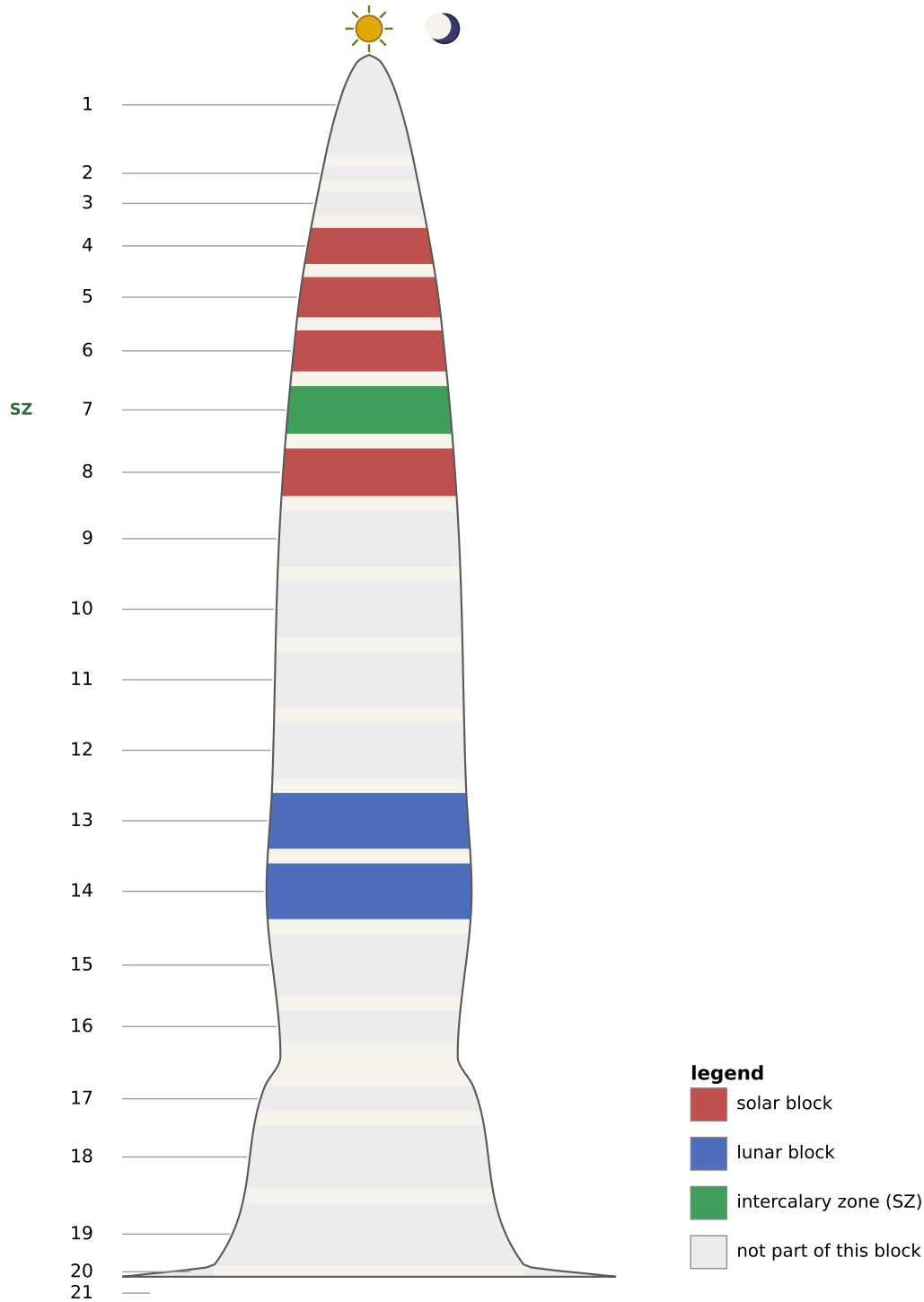
61.3 1.5 years / 18 lunations (6 QY / 18 L)



solar 510 d (zones 10-15) | lunar 492 d (zones 17-20) | delta 18 d

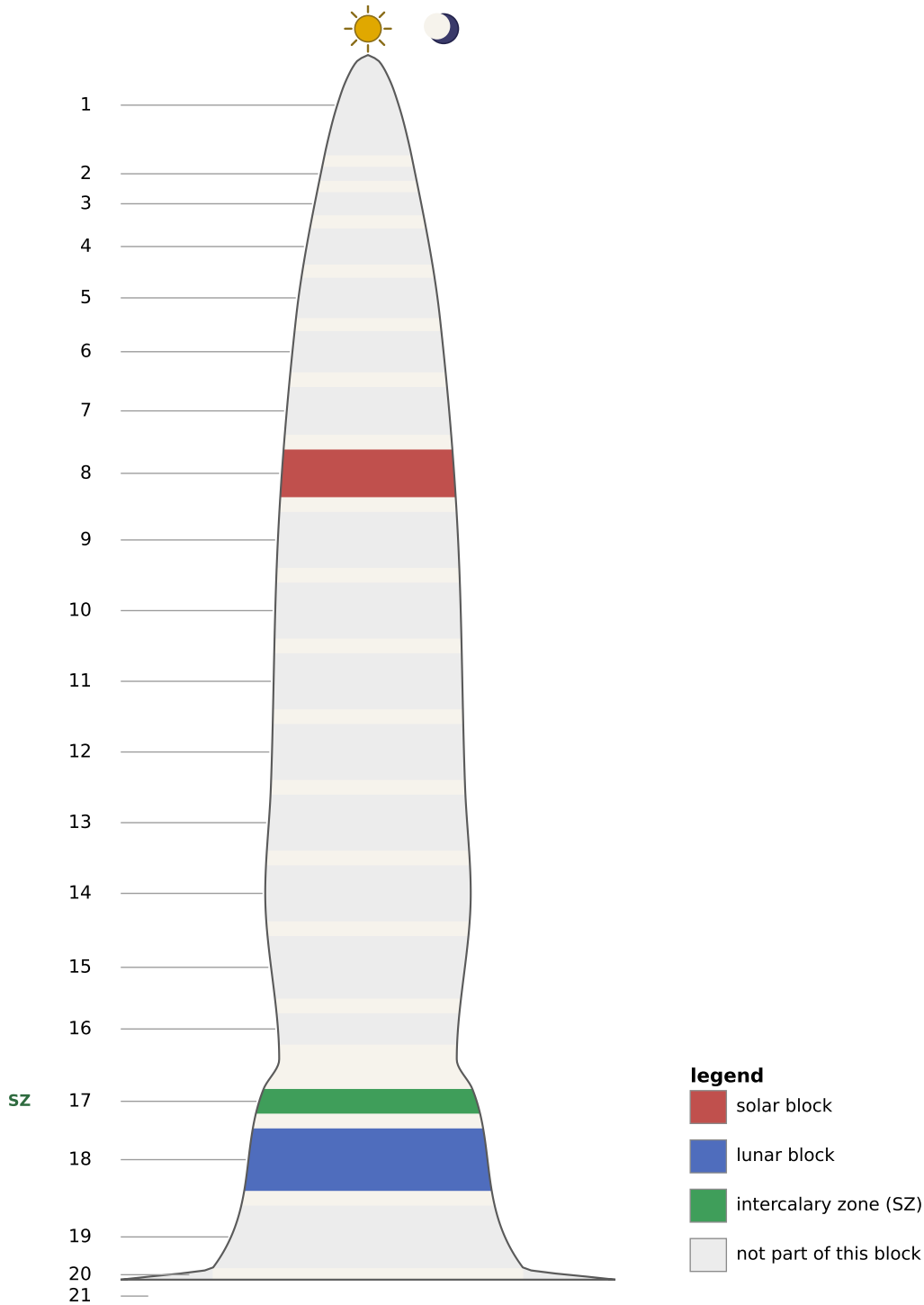
source: data/scenarios.json + data/zones.json + data/hat_parameters.json · generated by scripts/render_scenarios.py

61.4 1.75 years / 21 lunations (7 QY / 21 L)



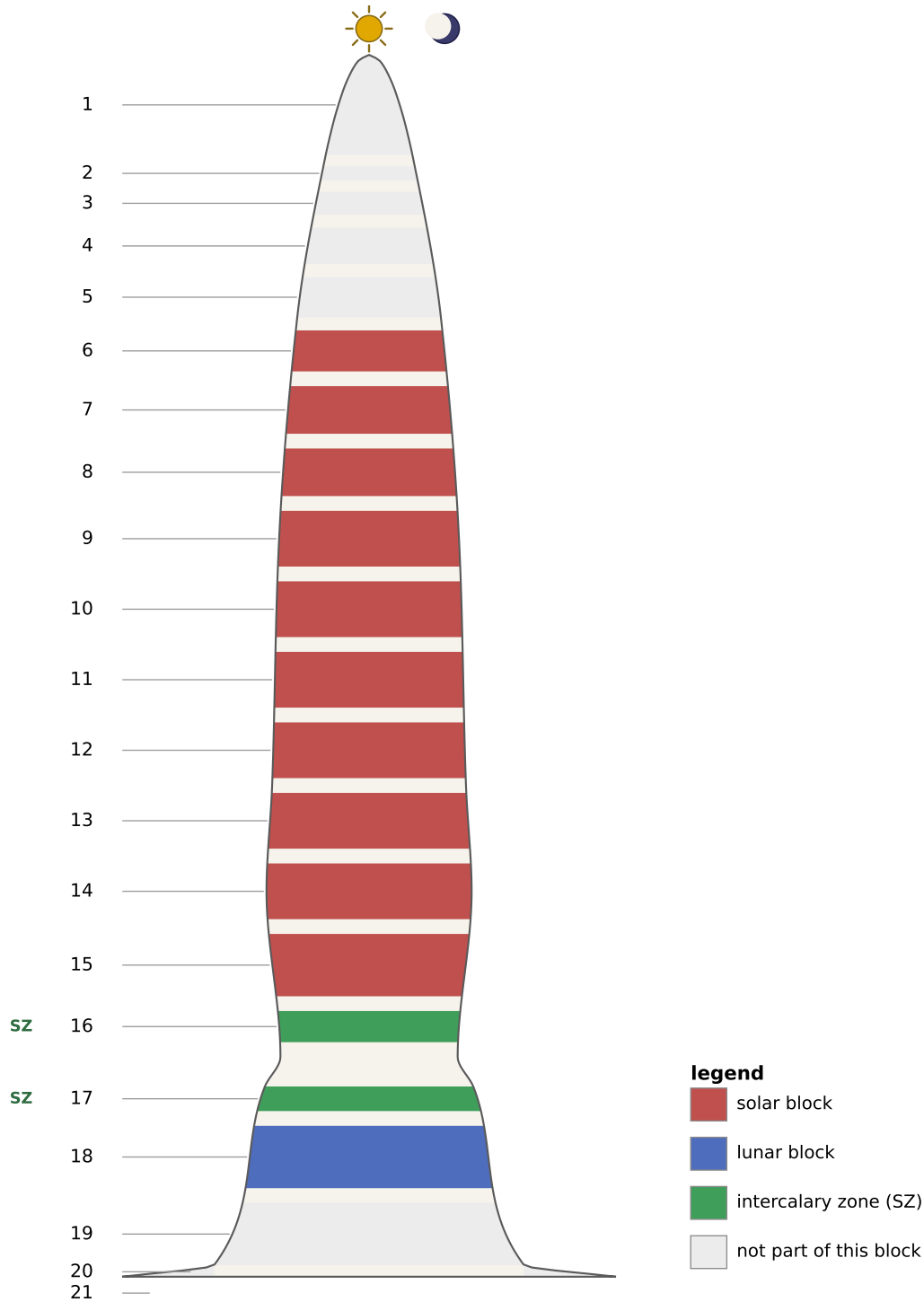
solar 146 d (zones 4-6 minus 7) | lunar 165 d (zones 13-14) | delta 19 d

62.1 2 years / 24 lunations (8 QY / 24 L)



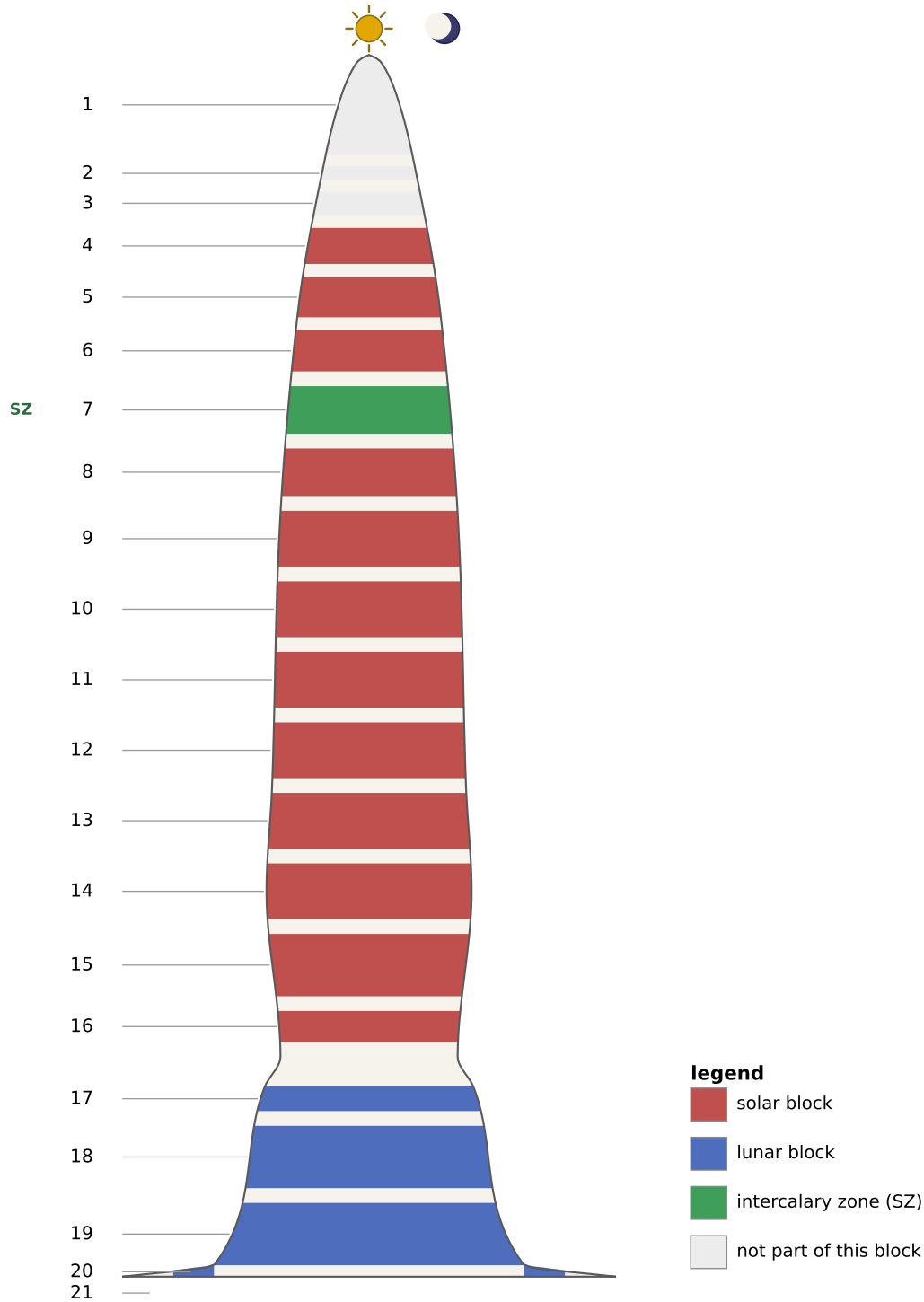
solar 105 d (zones 8) | lunar 126 d (zones 18) | delta 21 d

62.2 2.25 years / 27 lunations (9 QY / 27 L)



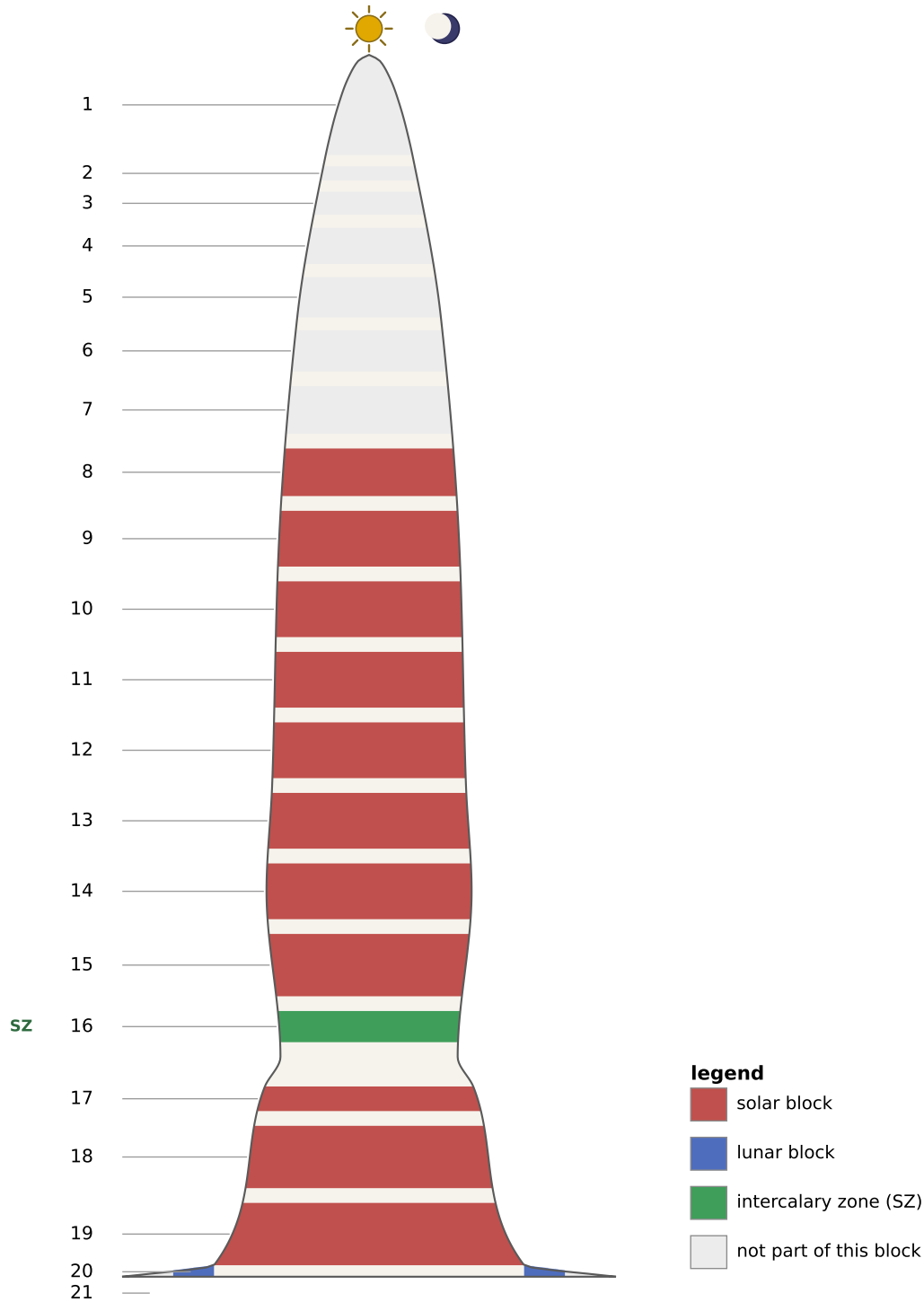
solar 252 d (zones 6-8) | lunar 230 d (zones 16-18) | delta 22 d

62.3 2.5 years / 30 lunations (10 QY / 30 L)



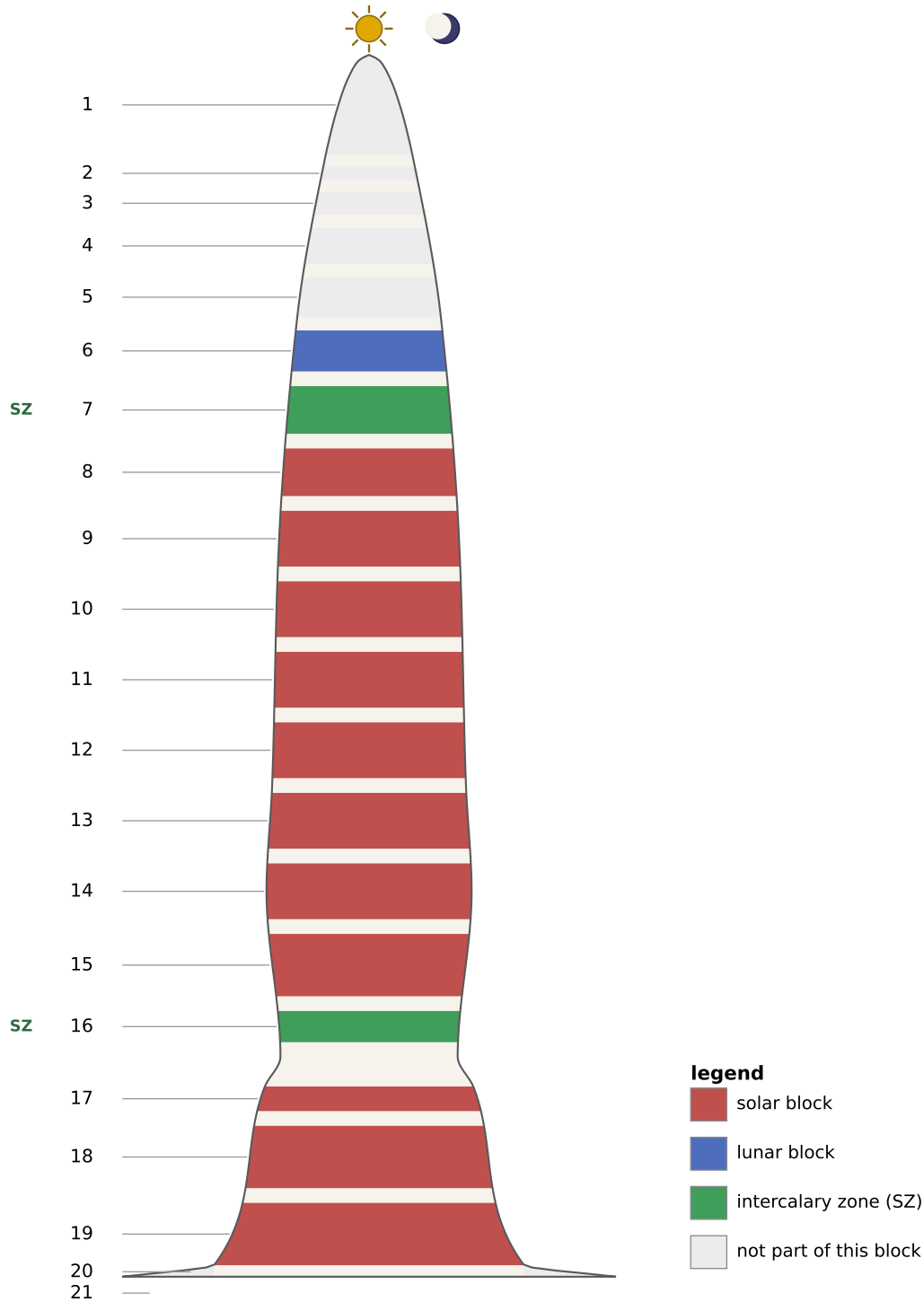
solar 913 d (zones 4-16 minus 7) | lunar 885 d (zones 12-20) | delta 28 d | INTERCALATION

62.4 2.75 years / 34 lunations (11 QY / 34 L)



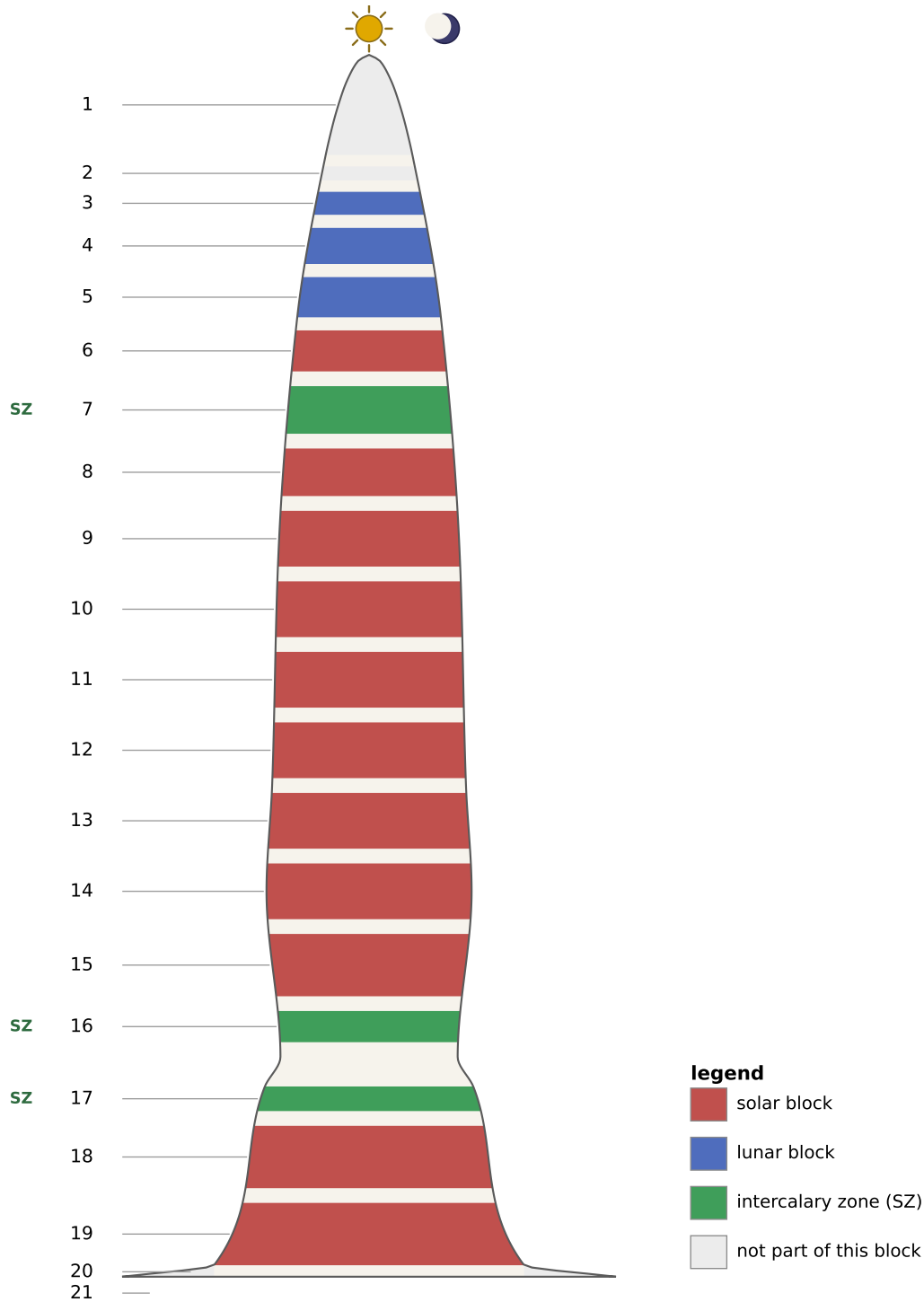
solar 1002 d (zones 8-19 minus 16) | lunar 1002 d (zones 10-20 minus 16) | delta 0 d

63.1 3 years / 37 lunations (12 QY / 37 L)



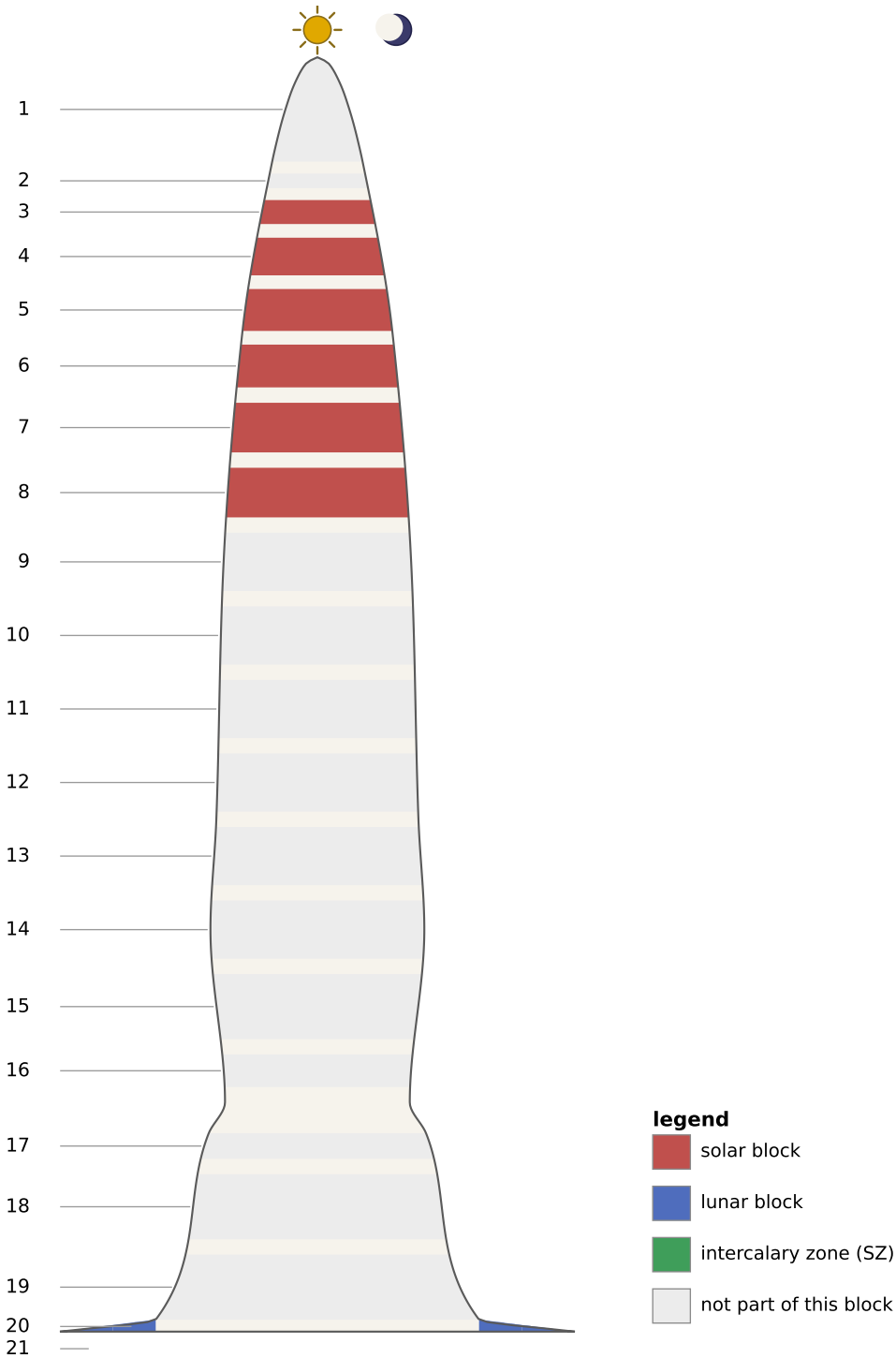
solar 1097 d (zones 7-19) | lunar 1092 d (zones 6-19 minus 7, 16) | delta 5 d

63.2 3.25 years / 40 lunations (13 QY / 40 L)



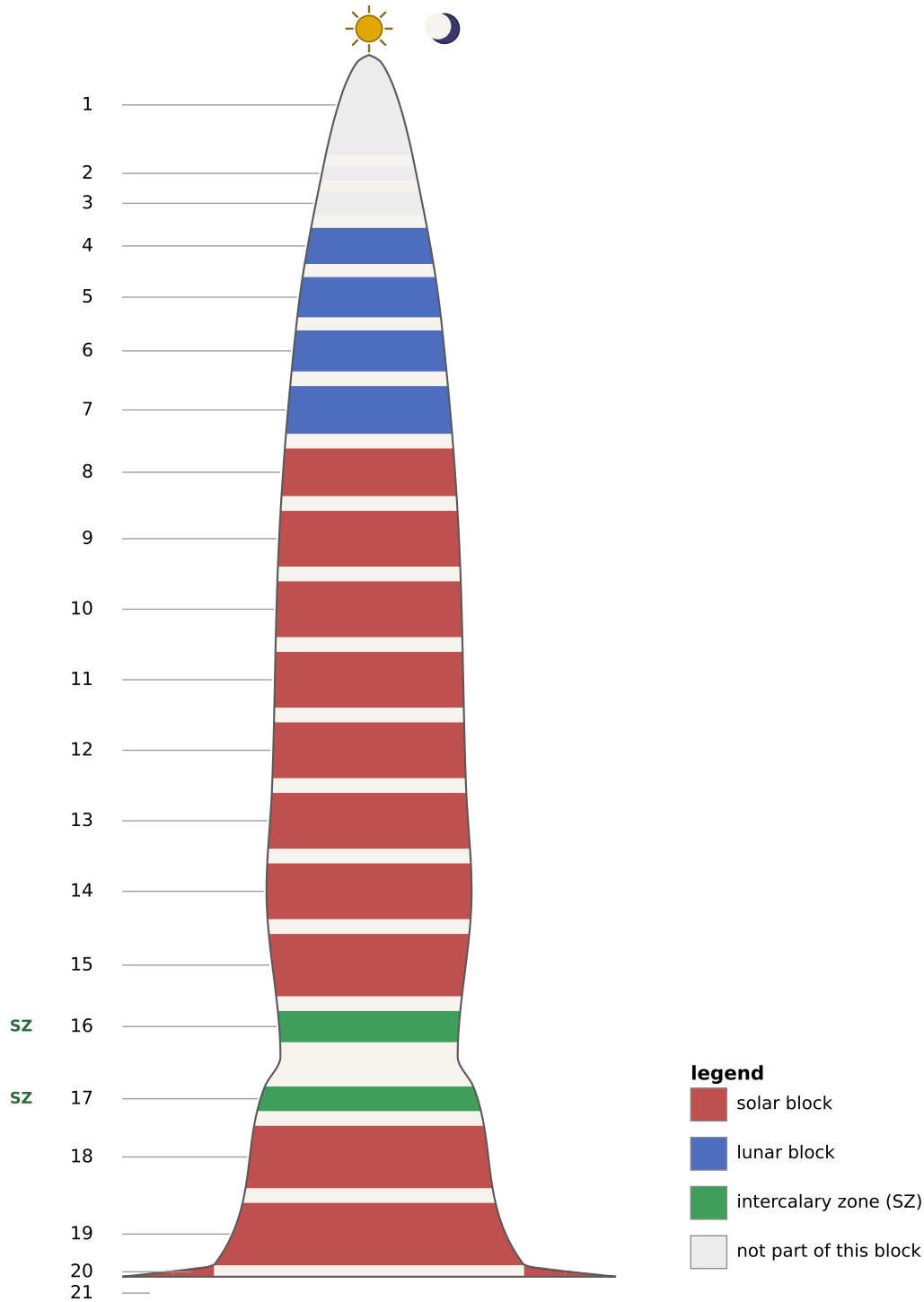
solar 1187 d (zones 6-19) | lunar 1181 d (zones 3-19 minus 7, 16, 17) | delta 6 d

63.3 3.5 years / 43 lunations (14 QY / 43 L)



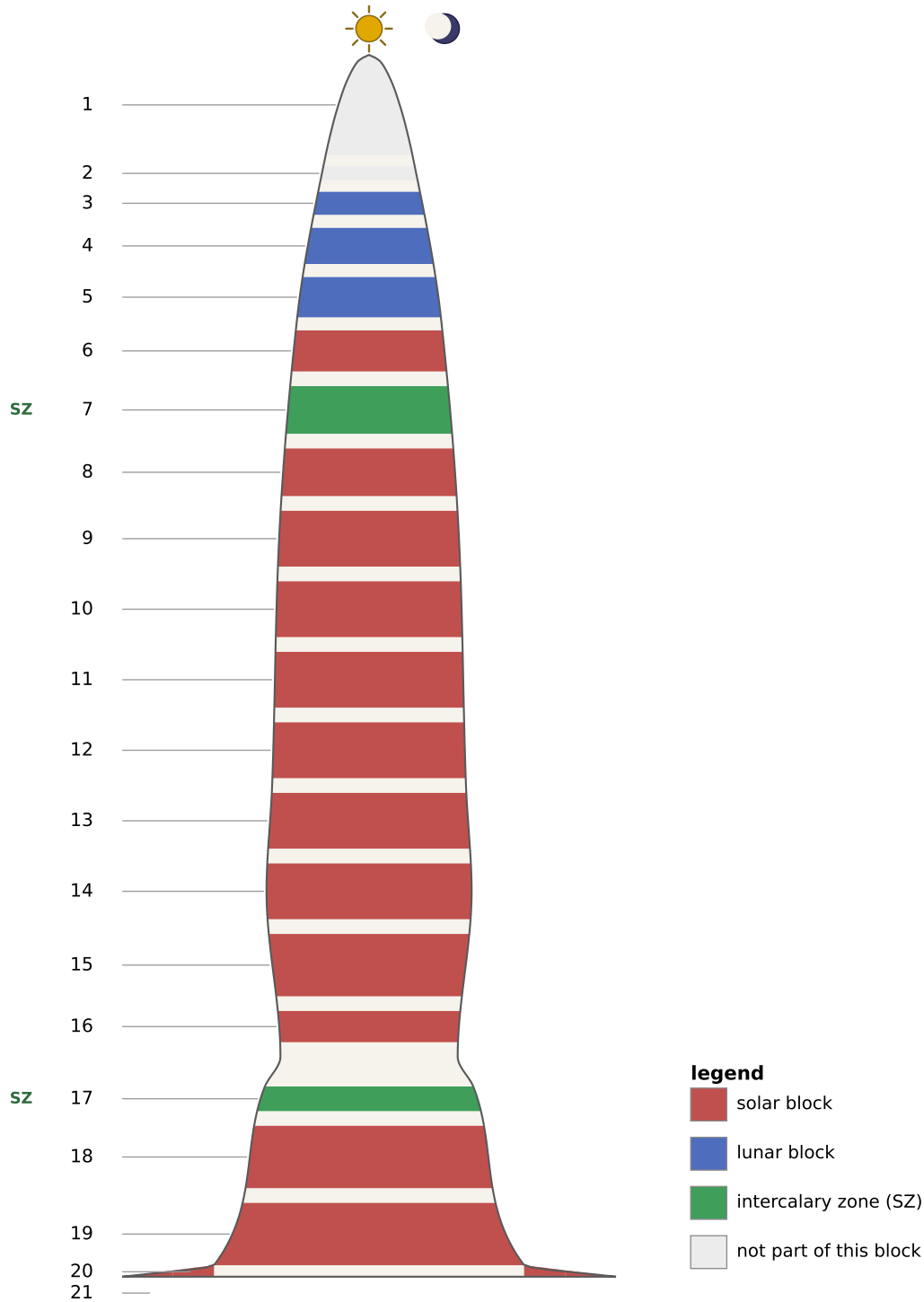
solar 407 d (zones 3-8) | lunar 397 d (zones 20-21) | delta 10 d

63.4 3.75 years / 46 lunations (15 QY / 46 L)



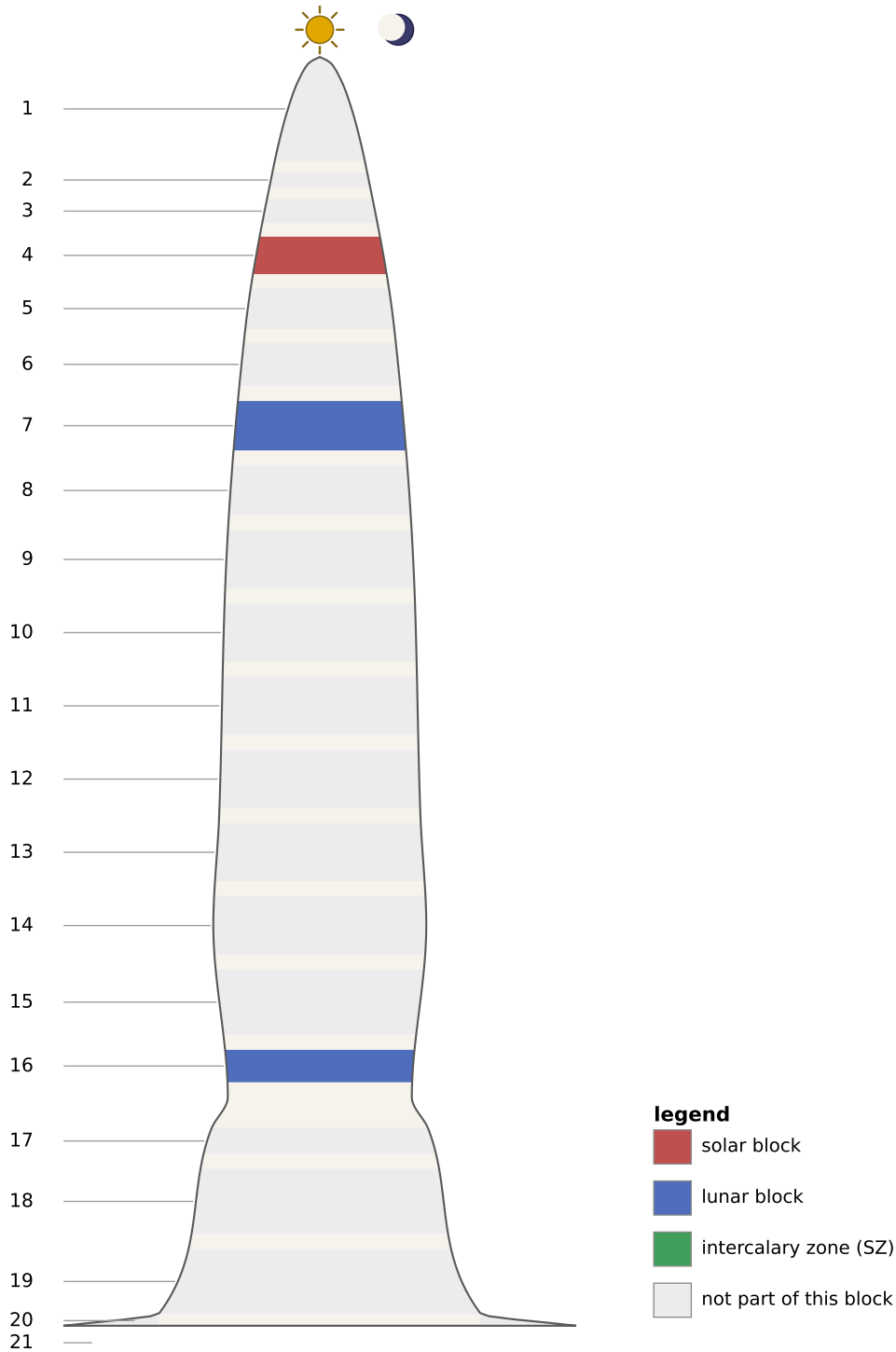
solar 1371 d (zones 8-21 minus 17) | lunar 1358 d (zones 4-20 minus 16, 17) | delta 13 d

64.1 4 years / 49 lunations (16 QY / 49 L)



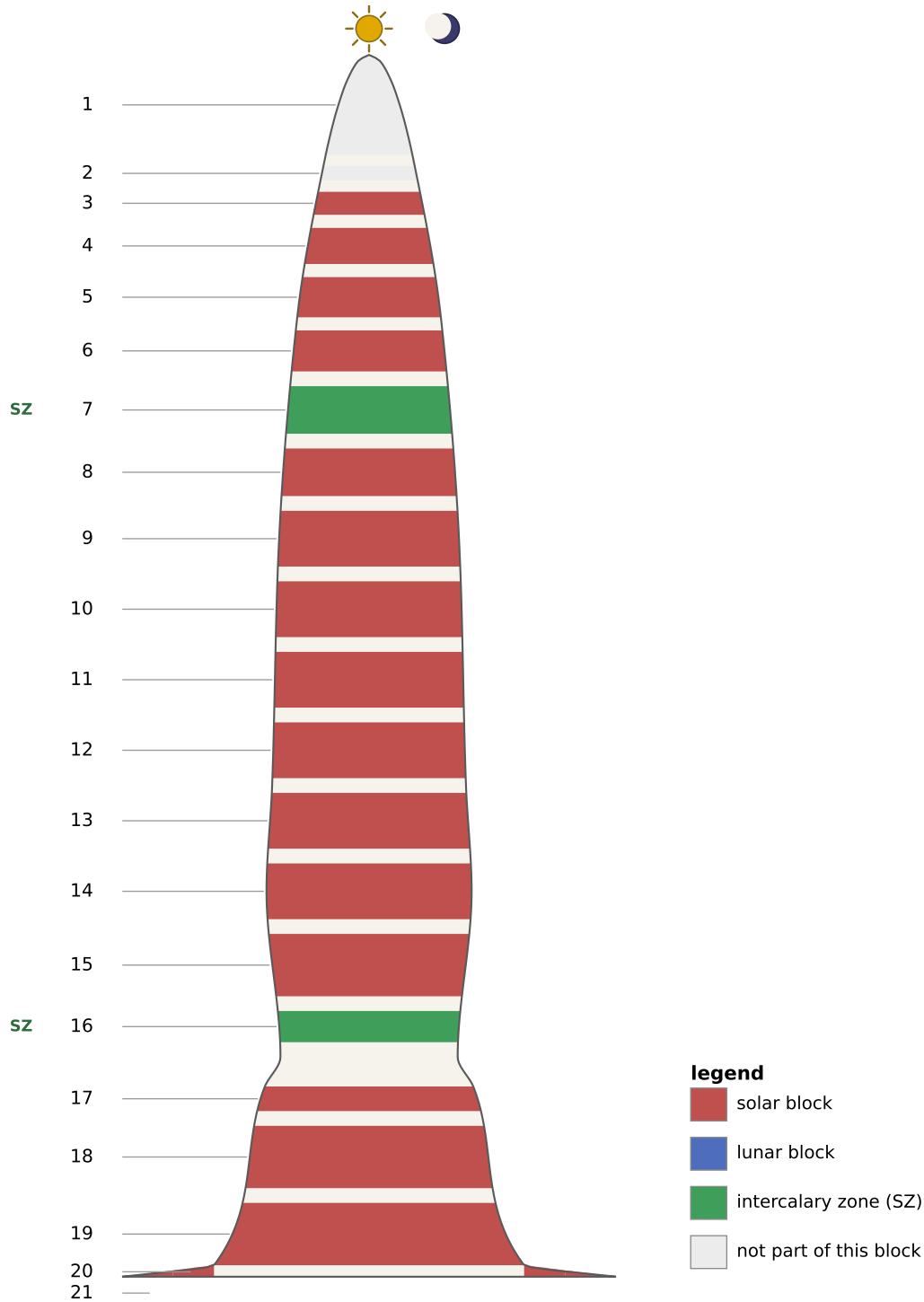
solar 1461 d (zones 6-21 minus 7, 17) | lunar 1447 d (zones 3-20 minus 7)
| delta 14 d

64.2 4.25 years / 52 lunations (17 QY / 52 L)



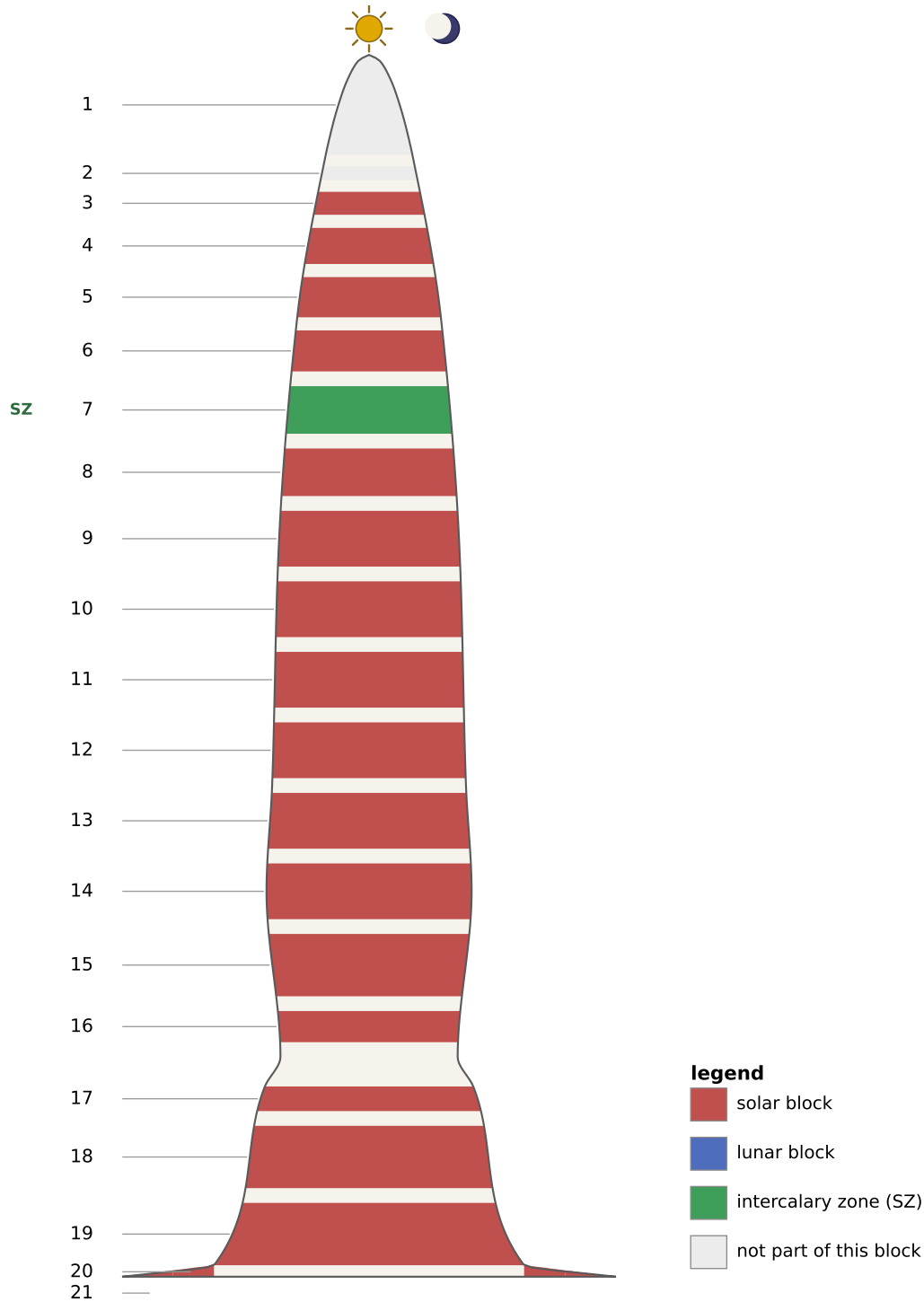
solar 75 d (zones 4) | lunar 95 d (zones 7, 16) | delta 20 d

64.3 4.5 years / 55 lunations (18 QY / 55 L)



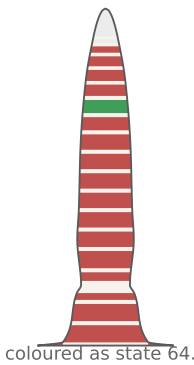
solar 1644 d (zones 3-21 minus 7, 16) | lunar 1622 d (zones 5-21) | delta 22 d | INTERCALATION

64.4 4.75 years / 59 lunations (full Metonic quarter) (19 QY / 59 L)



solar 1739 d (zones 3-21) | lunar 1739 d (zones 3-21) | delta 0 d

Four readings of the same object — units, targets and how much to believe



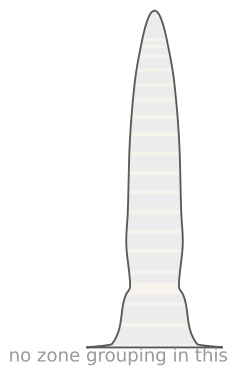
coloured as state 64.4

Menghin (2010) - lunisolar calendar

unit: days (rings x symbols) (total 1783)
target periods: 354 d, 365 d, 1739 d, 1742 d
confidence: hypothesis (provisional)

The strong 19-year claim is ~20-30%; the intercalary set moves between publications ({5,7,16,17} in 2000, {7,16,17} in 2010). See research/theories/menghin_lunisolar.md.

source: assets/menghin/2010/, pp. 47-63



no zone grouping in this reading

Schmidt-Kaler (2012) - Venus synodic cycle

unit: ornaments (all punched motifs) (total 651)
target periods: 292 d, 584 d, 2920 d
confidence: hypothesis, unproven here

Reuses Menghin's zone counts; the reading itself defines no zone colouring, and the ornament tally is NOT a day count. Not to be summed with Menghin's 1783.

source: research/sources/schmidt_kaler_2012.md



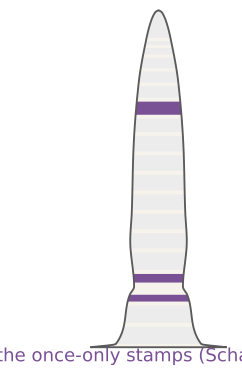
the 7 five-ring friezes

Lorenz - the structure is the Metonic cycle

unit: line-bands / friezes (total 235)
target periods: 235 d
confidence: re-reading of Menghin's counts

The 19 holds only as disc-bearing bands (measured: 21 zones); the 235 is arithmetic (12x12 + 7x13 for any 12+7 split), so it is not independent evidence.

source: research/sources/lorenz_2011.md



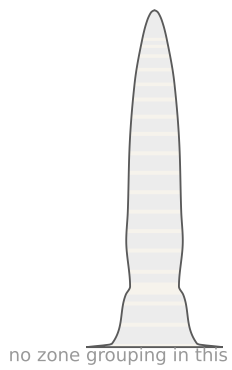
the once-only stamps (Schaltzonen)

Fuls (2004/2008) - pattern-type correction cycles

unit: rings x symbols, grouped by shared stamp
target periods: 945 d, 44415 d
confidence: candidate, refuted as selective on Berlin

The arithmetic is real and reproducible, but Fuls's own combinatorial null shows Berlin's hit-rate is indistinguishable from chance; only Schifferstadt is selective. 945 d = 15x59 + 60.

source: research/sources/fuls_2004.md



no zone grouping in this reading

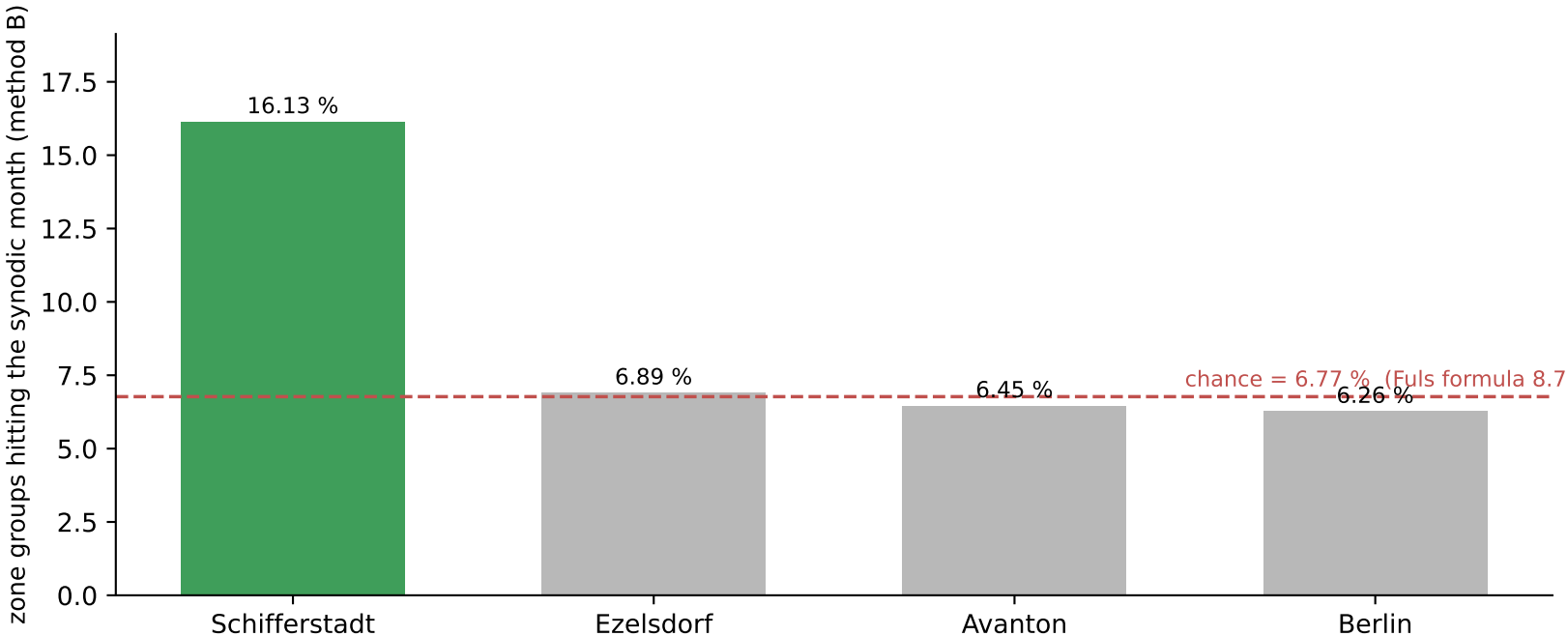
Sceptical - ornament, not a demonstrated calendar

unit: -
target periods: none
confidence: the null result

The numbers are easy to hit: free choice of blocks and of intercalary zones makes round targets appear at ~chance rate; the unit is not selective (counting plain symbols hits more often than counting rings).

source: research/theories/sceptical_ornament.md

Is the calendar selective? Fuls (2008) null model, re-run in-repo



Only Schifferstadt is clearly above chance. Berlin sits on the line, and Berlin's reading does not transfer to the other hats — so the calendar is a hypothesis, not a demonstrated function of the object.